Architectures of Expert Decision-Making

From Symbolic Logic to Strategic Agents: The Evolution of Domain-Specific Intelligence and Computational Expertise.

Foundational AI Structures

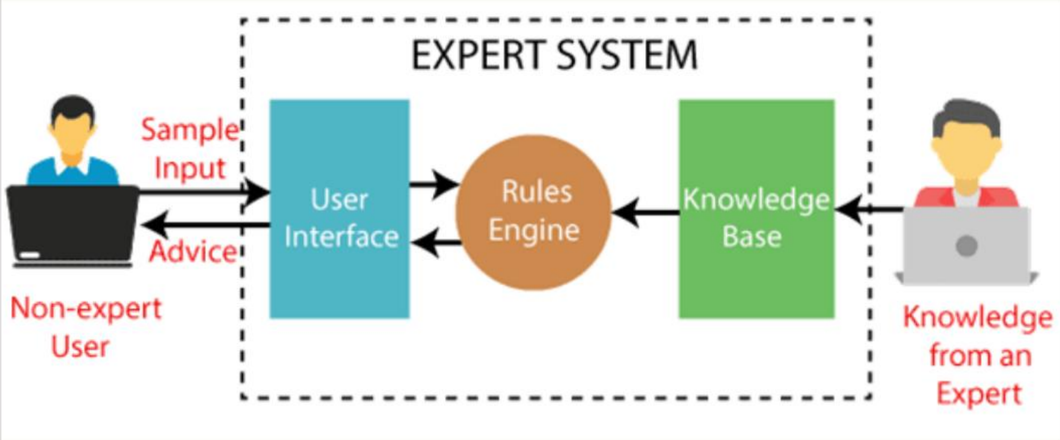
Exploring the Knowledge Base and Inference Engine Interface.

Core Expert System Components

Bifurcated Architecture

Expert systems simulate human specialists through a dual-component framework:

- **Knowledge Base:** The specialized repository of facts and heuristic rules (Symbolic IF-THEN logic).
- **Inference Engine:** The reasoning brain that processes knowledge to derive conclusions.



Inference Engine Chaining Strategies

Forward Chaining

Data-Driven Approach: Starts with known facts and applies rules to infer new info. Asks "What happens next?"

- Breadth-first search methodology.
- Ideal for real-time monitoring.
- Examples: Fraud detection, fire alarms.

Backward Chaining

Goal-Driven Approach: Starts with a hypothesis and works backward to find supporting facts.

- Depth-first search with backtracking.
- Efficient for diagnostic domains.
- Examples: Medical troubleshooting.

Logical Conflict Resolution

Specificity

Favoring rules with more preconditions to catch special cases.

Recency

Prioritizing rules matching the most recently added facts.

Priority

Using numerical ranks assigned by human experts to rules.

Refraction

Preventing rules from firing twice on the same facts.

Rule Evaluation Performance Metrics

- **Predictive Accuracy:** The system's fundamental ability to correctly reproduce expert conclusions from validated training datasets.
- **Coverage:** The breadth of the rule base; specifically, the proportion of possible domain cases for which the system has applicable logic.
- **Precision:** The statistical probability that a positive prediction made by a specific symbolic rule is empirically correct.

Association and Validation Metrics

Metric	Formula	Interpretation
Support	$\text{Supp}(X \cup Y) = \frac{\text{count}(X \cup Y)}{\text{Total Cases}}$	Frequency of the rule's itemset in data.
Confidence	$\text{Conf}(X \rightarrow Y) = \frac{\text{Supp}(X \cup Y)}{\text{Supp}(X)}$	Reliability of the implication logic.
Lift	$\text{Lift}(X \rightarrow Y) = \frac{\text{Supp}(X \cup Y)}{\text{Supp}(X) \times \text{Supp}(Y)}$	Strength of association (>1 is positive).

Knowledge Acquisition via LAPS

Dechunking Expertise

LAPS (Logic Analysis and Problem Solving) is a specialized interview methodology that interweaves acquisition and validation.

By using **"dechunking"**, the process refines sample solution paths to map an expert's internal reasoning, effectively closing logic gaps before system implementation.

Fuzzy Logic: Reasoning with Vague Data

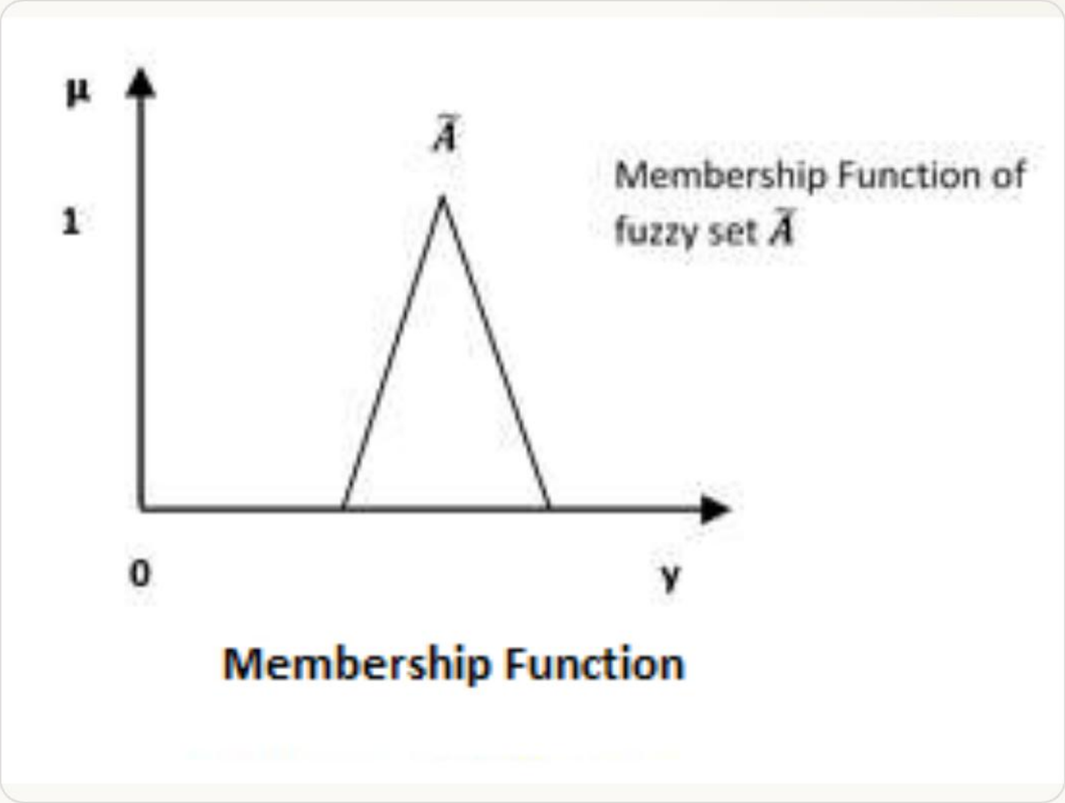
Handling Imprecision

Fuzzy systems move beyond binary True/False logic to handle human-centric concepts like "warm" or "fast".

Fuzzification: Mapping crisp values (70°F) to fuzzy sets (0.8 Warm).

Membership Functions: Defining degrees of membership between 0 and 1.

Defuzzification: Converting fuzzy output back to crisp action (e.g., Centroid Method).



Fuzzy Certainty: Zadeh Operators

AND (Intersection)

Used to combine preconditions where the minimum certainty level limits the outcome.

$$cf(P1 \text{ AND } P2) = \min(cf(P1), cf(P2))$$

OR (Union)

Used for alternate paths where the strongest certainty level dictates the outcome.

$$cf(P1 \text{ OR } P2) = \max(cf(P1), cf(P2))$$

Case Study: Magic Square Transformation

Reducing Pattern Matching to Arithmetic

By mapping Tic-Tac-Toe squares to a 3x3 Magic Square, winning patterns are simplified to any three numbers summing to 15.

8	1	6
3	5	7
4	9	2

Perfect Play: Tic-Tac-Toe Rules

1. **Win:** Complete a row of three.
2. **Block:** Block opponent's win.
3. **Center:** Take center if available.
4. **Opposite:** Take opposite corner.
5. **Corner:** Take any empty corner.
6. **Side:** Take any empty side.
7. **Catch-all:** Random available spot.

The Challenge of Chess

From Intractable to Strategic

Unlike simple games, Chess is computationally **intractable**. Scaling expert systems to this domain requires Bitboards and adversarial search.

Deep Blue utilized Minimax with **Alpha-Beta Pruning** and hand-crafted evaluation functions tuned by Grandmasters, representing the pinnacle of symbolic AI.

Expert System Implementation Roadmap

Acquisition

Decompile knowledge via LAPS interviews.

Inference

Choose Forward vs Backward chaining.

Validation

Apply Support, Confidence, & Lift metrics.

Uncertainty

Model vague data with Fuzzy Logic.

Questions & Discussion

Architectures of Expert Decision-Making: Bridging Symbolic Logic and Strategic Intelligence.

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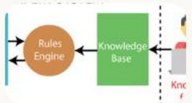
Image Sources



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